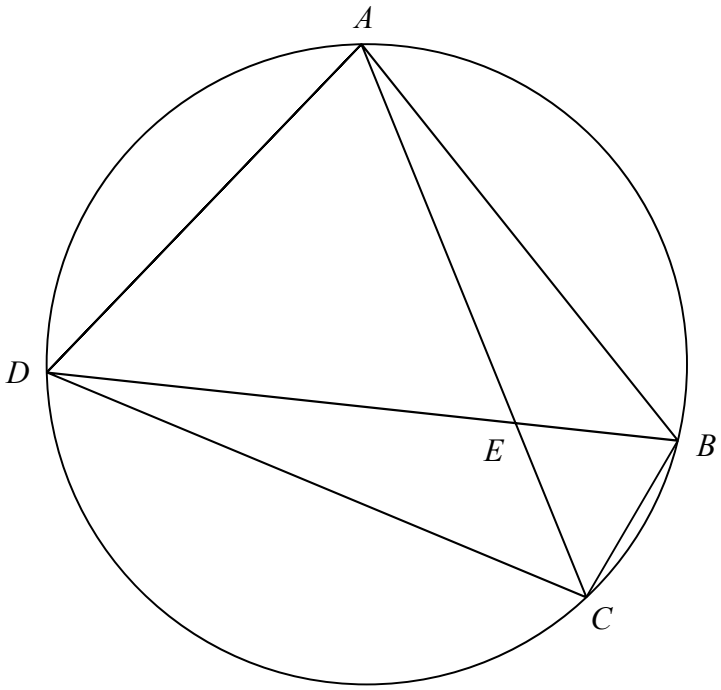
22 A, B, C and D are four points on a circle.



AEC and DEB are straight lines.

Triangle AED is an equilateral triangle.

Prove that triangle ABC is congruent to triangle DCB.

AED is equilateral: angle DAE = angle ADE = 60°.

Angle ACB = angle ADB = 60° (same chord AB).

Angle DBC = angle DAC = 60° (same chord DC).

Angle ABD = angle ACD (same chord AD).

Hence angle ABC = angle ABD + 60°

and angle DCB = angle ACD + 60°: these are equal.

BC is common. The two triangles are congruent by ASA.

1 mark
Equal-angle pair with reason

1 mark
Both 60° angles with reasons

1 mark
Second angle pair

1 mark
Common BC; ASA

(Total for Question 22 is 4 marks)